On the Adaptive Performance Control of Robot Manipulators

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***Abstract*— In this paper, we propose an adaptive performance control (APC) framework for output-feedback control of robot manipulators with input constraints, focusing on achieving joint position tracking with adaptive performance attributes. Building on the APC methodology, we introduce a single-funnel control approach that dynamically adjusts performance constraints to accommodate the actuator limitations inherent in robotic systems. Moreover, a robust tracking differentiator with predefined convergence characteristics is developed to estimate the unmeasured velocity errors. The effectiveness of the proposed control scheme is validated through simulation results on a 2-DoF planar manipulator and experiments on a 3-fingered 12 degree-of-freedom robotic hand tasked with performing a musical piece on a keyboard.**

I. Introduction

Control of robot manipulators represents one of the leading streams of robotics research, driven by ever-growing demands for precision, adaptability, and efficiency coming from industrial, medical, and service sectors. A robot manipulator is generally an articulated mechanical system with interconnected links and joints that, under complex coordinated movements, are able to carry out tasks of precision such as assembly, welding, and handling in dynamic environments. However, the nonlinear dynamics, joint couplings and uncertainties in both system parameters and operating environment make the manipulator control strategy design difficult.

Classical approaches of joint position tracking includes computed torque, inverse dynamics, and passivity-based control have predominantly relied on detailed dynamic models to control joint position trajectories [1]. While effective, these model-based approaches often face limitations due to inevitable modeling inaccuracies, which can compromise tracking accuracy and velocity performance [2]–[4]. To address these challenges, model-free controllers, such as linear PD and PID control structures, have been explored to achieve position tracking without a model dependency, although their gains require careful tuning to ensure adequate tracking performance [5].

Recently, structurally simple controllers have been proposed to meet prescribed performance specifications for joint position tracking by employing techniques like prescribed performance control (PPC), originally designed for uncertain MIMO nonlinear systems [6], and funnel control [7]. Such approaches achieve predefined transient and steady-state performance for a combined error of position and velocity tracking. The authors in [8] developed a controller for joint position and velocity tracking with prescribed performance guarantees, eliminating the need for dynamic models but lacking input constraints. A PPC approach for nonlinear systems with actuator saturation, developed in [9], presenting a control method that adjusts dynamically based on actuator saturation, though limited to single-input single-output (SISO) class of systems. An observer-based PPC controller with application to a robotic manipulator was recently presented in [10]. An extension of the PPC technique, called adaptive performance control (APC) [11], [12] has emerged as a robust method for managing both input and output constraints in unknown nonlinear systems. APC dynamically adjusts soft performance constraints in real time to address the strict input limitations imposed by actuators of the system.

In this work, we build on the APC methodology, proposing a single-funnel output-feedback control scheme that ensures prescribed performance specifications for the joint position tracking errors in an n degree-of-freedom (DoF) robotic manipulator under hard input constraints. The proposed approach leverages a novel tracking differentiator (TD) to estimate the velocity error directly from the position error measurements, which is particularly advantageous in applications where only position feedback is available and reference velocity is unknown. The incorporation of the TD enables the controller to achieve high tracking accuracy with reduced gain requirements, thus mitigating the impact of noise without relying on any knowledge of the robot's dynamical model or learning structures. The effectiveness of the proposed controller is validated through comprehensive simulations on a 2-DoF planar manipulator, as well as via experiments on a 3-fingered 12-DoF robotic hand tasked with tracking a reference trajectory to perform a musical piece on a keyboard.

II. Problem Statement

The dynamical model of an n-DoF robotic manipulator is governed by the Euler-Lagrange equation:

$$H(q)\ddot{q} + C(q,\dot{q})\dot{q} + G(q) + F(t) = u \qquad (1)$$

where $q, \dot{q}, \ddot{q} \in \mathbb{R}^n$ denote the joint position, velocity, and acceleration vectors, respectively. Additionally, $H(q) \in \mathbb{R}^{n\times n}$ is the positive definite inertia matrix, $C(q,\dot{q})\dot{q} \in \mathbb{R}^n$

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denotes the centripetal and Coriolis forces, $G(q) \in \mathbb{R}^n$ is the gravity vector and $F(t) \in \mathbb{R}^n$ captures bounded disturbances. The control input $u \in \mathcal{U} \subset \mathbb{R}^n$, applied through the actuation mechanism, i.e., motors, is absolutely bounded by the known positive constants $\bar{u}_i$ for each joint $i = 1, \ldots, n$, due to physical constraints. These input constraints define the feasible control set $\mathcal{U} \coloneqq \{u \in \mathbb{R}^n : |u_i| \leq \bar{u}_i,\ i = 1, \ldots, n\}$.

Consider a desired joint position vector $q_d(t) \in \mathbb{R}^n$, which is a bounded, twice differentiable function of time with unknown but bounded derivatives $\dot{q}_d$ and $\ddot{q}_d$. Additionally, define the output tracking error as $e(t) \coloneqq q(t) - q_d(t) = [q_1(t) - q_{d_1}(t), \ldots, q_n(t) - q_{d_n}(t)]^T \in \mathbb{R}^n$.

The objective of this work is to design an output-feedback controller for the robot (1) that enforces adaptive transient and steady-state performance specifications on $e(t)$. The performance specifications should dynamically adjust to accommodate the input constraints imposed by the actuation limitations of the plant, ensuring effective output tracking despite the presence of input constraints.

The following assumptions are established:

Assumption 1: Only the output tracking error $e(t)$ is available for measurement. However, a known compact set $\mathcal{E}_0$ exists such that $[e^T(0), \dot{e}^T(0)]^T \in \mathcal{E}_0$.

Assumption 2: The model parameters $H(q), C(q, \dot{q}), g(q)$ and the disturbance function $F(t)$ are completely unknown.

Remark 1: Assumption 1 allows for scenarios where the desired trajectory is not analytically known, but instead measured in real-time, e.g. target tracking applications.

III. Robust Tracking Differentiator

Here, a robust TD is introduced to estimate unmeasured velocity errors using only position data. The proposed design ensures convergence within predefined bounds, allowing for accurate velocity estimation essential for effective feedback control. Let us first define the deadzone function:

$$s_\omega(\chi) \coloneqq \begin{cases} 0 & \text{if } |\chi| < \omega \\ \frac{|\chi| - \omega}{1 - |\chi|} & \text{if } \omega \leq \chi < 1 \end{cases},\ \omega \in (0, 1). \tag{2}$$

The TD is then given by:

$$\begin{aligned} \dot{\epsilon}_{1,i} &\coloneqq \epsilon_{2,i}(t) \\ \dot{\epsilon}_{2,i} &\coloneqq -\gamma_1 a_i^2(t)\left(\epsilon_{1,i}(t) - e_i(t)\right) - \gamma_2 a_i(t)\epsilon_{2,i}(t) \end{aligned} \tag{3}$$

where $\gamma_1,\ \gamma_2$ denote constant positive gains and the adaptive gain $a_i(t)$ is computed by:

$$a_i(t) \coloneqq s_\omega(\xi_i(t)) + \int_0^t s_\omega(\xi_i(\tau))d\tau \tag{4}$$

with $\xi_i(t) \coloneqq \frac{\epsilon_{1,i}(t) - e_i(t)}{\rho_{o_i}(t)}$ where $\rho_{o_i}(t) \coloneqq (\rho_{o_i}(0) - \rho_{o_i}^f)e^{-l_{o_i}t} + \rho_{o_i}^f$ with $\rho_{o_i}(0) > |\epsilon_{1,i}(0) - e_i(0)|$ and $l_{o_i}, \rho_{o_i}^f$ for $i = 1, \ldots, n$ denoting positive constants encapsulating the desired transient and steady state performance specifications of the proposed TD.

Theorem 1: Consider a bounded and continuous reference signal $e_i : \mathbb{R}^+ \to \mathbb{R}$, with an unknown but bounded first derivative. The TD (3) is exponentially convergent with rate l_{o_i} and accuracy level $\rho_{o_i}^f$, meaning that the estimation errors $\epsilon_{1,i}(t) - e_i(t)$ and $\epsilon_{2,i}(t) - \dot{e}_i(t)$ converge to neighborhoods of the origin of order $O(\rho_{o_i}^f)$ at a rate faster than $e^{-l_{o_i}t}$, $i = 1, \ldots, n$.

Proof: The proof follows three steps.

Step 1. We begin by establishing the existence of a unique maximal solution $[\xi_i(t), \epsilon_{1,i}(t), \epsilon_{2,i}(t)]^T$ over the domain $(-1, 1) \times \mathbb{R}^2$ within the time interval $[0, t_{\max})$ with $t_{max} \in \{\mathbb{R}^*_{>0}, \infty\}$. Specifically, since $\rho_{o_i}(0) > |\epsilon_{1,i}(0) - e_i(0)|$, it follows that $\xi_i(0) \in (-1, 1)$. Furthermore, the dynamics (3) is continuous and locally integrable with respect to t, and locally Lipschitz in $\xi_i, \epsilon_{1,i}(t), \epsilon_{2,i}(t)$. Therefore, by applying Theorem 54 from [13] (pp. 476), we establish the existence and uniqueness of a maximal solution $[\xi_i(t), \epsilon_{1,i}(t), \epsilon_{2,i}(t)]^T$ over $(-1, 1) \times \mathbb{R}^2$ for all $t \in [0, t_{\max})$.

Step 2. We then proceed to demonstrate by contradiction that $\xi_i(t) \in (-1, 1)$ for all $t \geq 0$, thereby ensuring that the solution $\epsilon_{1,i}(t), \epsilon_{2,i}(t)$ of the TD (3) is extended indefinitely. Since $a_i(t)$ is strictly positive for all t, the real parts of the eigenvalues of the pseudo-linear system (3) are negative, leading to bounded nonlinear eigenvectors. This implies that the state $\epsilon_{1,i}(t), \epsilon_{2,i}(t)$ remains bounded when $e_i(t)$ and $\dot{e}_i(t)$ are also bounded. Additionally, when $\xi_i(t) \to \pm 1$, $a_i(t)$ would diverge to $+\infty$. However, by Theorem 1 from [14], $\xi_i(t)$ converges uniformly to zero over time, which contradicts $\xi_i(t) \to \pm 1$. Consequently, $a_i(t)$ remains bounded, ensuring $\xi_i(t) \in (-1, 1)$ for all $t \geq 0$.

Step 3. Applying Theorem 1 from [14], there exists a bound $\bar{a}_i$ such that $a_i(t) \geq \bar{a}_i$ guarantees exponential convergence of the estimation errors $\epsilon_{1,i}(t) - e_i(t)$ and $\epsilon_{2,i}(t) - \dot{e}_i(t)$ at rate l_{o_i} to a neighborhood $O(\rho_{o_i}^f)$ around the origin. Specifically, the adaptive parameter $a_i(t)$ eventually grows to exceed $\bar{a}_i$, ensuring that the estimation errors remain within the deadzone regions $|\epsilon_{1,i}(t) - e_i(t)| \leq \omega\rho_{o_i}(t)$ and $|\epsilon_{2,i}(t) - \dot{e}_i(t)| \leq O(\omega\rho_{o_i}(t))$, concluding the proof. ■

Remark 2: The TD in Eq. (3) is designed to estimate the derivatives of the tracking error enhancing the tracking performance of robot manipulators where only position measurements are available, and the reference velocity is not known beforehand. In robotic control, the absence of direct velocity or acceleration measurements presents a challenge, as precise tracking requires accurate velocity information. By employing the proposed TD algorithm, the system can approximate the derivative of the output error directly from position data, reducing the need for high-gain control actions commonly used to correct tracking errors. This approach allows for lower control gains, which, through feedback of the estimated error derivative, reduces sensitivity to sensor noise and external disturbances that often affect robotic systems with joint friction, backlash and environmental interferences.

IV. Controller Design

We, now, present the APC-based controller, incorporating the estimates generated by the TD (3). First, we define the saturation function:

$$\text{sat}_\mu(\chi) \coloneqq \begin{cases} \chi\text{sign}(\chi) & \text{if } |\chi| > \mu \\ \chi & \text{if } |\chi| \leq \mu. \end{cases}$$

By leveraging the estimates from (3) we define the tracking error surface:

$$\sigma(t) \coloneqq \mathrm{sat}_{\bar{\epsilon}_2}(\epsilon_2(t)) + Q\mathrm{sat}_{\bar{\epsilon}_1}(\epsilon_1(t)) \tag{5}$$

where $\mathrm{sat}_{\bar{\epsilon}_i}(\epsilon_i) \coloneqq [\mathrm{sat}_{\bar{\epsilon}_{i,1}}(\epsilon_{i,1}), \ldots, \mathrm{sat}_{\bar{\epsilon}_{i,n}}(\epsilon_{i,n})]^T \in \mathbb{R}^n$ with $\bar{\epsilon}_i \coloneqq [\bar{\epsilon}_{i,1}, \ldots, \bar{\epsilon}_{i,1}]^T \in \mathbb{R}^n$ for $i = 1, 2$. From this point onward, our approach imposes performance constraints on $\sigma(t)$, which has been shown in [15] to be sufficient for enforcing performance specifications on the output tracking error $e(t)$. The saturation of the error estimates is exploited to mitigate the peaking phenomenon induced by the TD, during the transient, which may degrade the overall performance of the system. The saturation levels in $\bar{\epsilon}_i$ should be lie outside of $\mathcal{E}_0$. Let us also define the norm $\sigma_n(t) \coloneqq \|\sigma(t)\| \in \mathbb{R}_{>0}$ and the unit vector $\sigma_u(t) \coloneqq \frac{\sigma(t)}{\sigma_n(t)}$. Then, we design the nominal control signal:

$$u_d(t) \coloneqq -K \frac{1}{2\left(1 - \frac{\sigma_n(t)}{\rho_u(t)}^2\right)} \ln\left(\frac{1 + \frac{\sigma_n(t)}{\rho_u(t)}}{1 - \frac{\sigma_n(t)}{\rho_u(t)}}\right) \sigma_u(t) \tag{6}$$

where $K = \mathrm{diag}(k_i),\ i = 1, \ldots, n$ denotes a diagonal gain matrix, with $k_i > 0$. The constrained control input is then given by:

$$u(t) \coloneqq \mathrm{sat}_{\bar{u}}\left(u_d(t)\right) \coloneqq \begin{bmatrix} \mathrm{sat}_{\bar{u}_1}(u_{d_1}(t)) \\ \vdots \\ \mathrm{sat}_{\bar{u}_n}(u_{d_n}(t)) \end{bmatrix} \in \mathcal{U}. \tag{7}$$

Finally, the adaptive law that provides a dynamic trade-off between input and output constraints is obtained by:

$$\dot{\rho}_u \coloneqq -l_u(\rho_u(t) - \rho_{u,\infty}) + k_u \frac{\sigma_u^T(t)(u(t) - u_d(t))}{\frac{\sigma_n(t)}{\rho_u(t)}} \tag{8}$$

where $l_u,\ \rho_{u,\infty}$ are positive parameters encapsulating the desired transient and steady state performance specifications, respectively, $k_u > 0$ is a constant gain and $\rho_u(0)$ is set such that $\rho_u(0) > |\sigma_n(0)|$. Note that the first term in (8) imposes the prescribed output performance specifications to the system, whereas the second term, which is non negative for all $t \geq 0$, relaxes the performance envelope by increasing $\rho_u(t)$ when the actuators reach saturation. This adjustment maintains the nominal control signal bounded, ensures that $\rho_u(t) \geq \rho_{u,\infty}$, and guarantees output tracking with adaptive performance based on the hard input constraints.

Theorem 2: Consider a robot manipulator governed by (1) and a smooth reference trajectory $q_d(t)$, satisfying Assumptions 1 and 2. The proposed output-feedback controller (5)-(8), fed by the error estimates of TD (3), ensures tracking of both $q_d(t)$ and $\dot{q}_d(t)$ with adaptive performance specifications and guarantees the boundedness of all closed-loop signals for all $t \geq 0$.

Proof: In what follows, we omit the time argument for the closed-loop signals in order to increase the clarity of the proof. Let us first write (1) in normal form as:

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= f(x) + g(x_1)u \end{aligned} \tag{9}$$

with $x_1 \coloneqq [q_1, \ldots, q_n]^T$, $x_2 \coloneqq [\dot{q}_1, \ldots, \dot{q}_n]^T$, $x \coloneqq [x_1^T, x_2^T]^T$, $f(x) \coloneqq -H^{-1}(x_1)[C(x)x_2 + G(x_1) + F]$ and $g(x_1) \coloneqq H^{-1}(x_1)$. Furthermore, let us define the normalized tracking error $\zeta \coloneqq \frac{\sigma_n}{\rho_u}$ and the closed-loop state vector $\chi \coloneqq [x^T, \zeta]^T \in \mathbb{R}^{2n} \times (-\infty, 1)$. Differentiating χ with respect to time we obtain:

$$\dot{\chi} \coloneqq \begin{bmatrix} x_2 \\ f(x) + g(x_1)u \\ \frac{\sigma_u^T \dot{\sigma} \rho_u - \sigma_n \dot{\rho}_u}{\rho_u^2} \end{bmatrix} \tag{10}$$

By the initial values of the performance function $\rho_u(0)$, it follows that $\zeta(0) < 1$, implying $\zeta(0) \in (-\infty, 1)$. Additionally, (10) is piecewise continuous and locally integrable on t, as well as locally Lipschitz on χ. Therefore, following the results from [13], a maximal solution $\chi(t)$ of (10) exists over a time interval $[0, t_{\max})$ such that $\chi(t) \in \mathbb{R}^{2n} \times (-\infty, 1)$ for all $t \in [0, t_{\max})$ with $t_{max} \in \{\mathbb{R}^*_{>0}, \infty\}$.

Next, it can be easily deduced that the system (9), satisfies the input-to-state stability property. Thus, owing to the input saturation, there exist a class $\mathcal{K}$ function $\alpha(\cdot)$ and a class $\mathcal{KL}$ function $\beta(\cdot,\cdot)$ such that $x \in \Omega_x$, where $\Omega_x \coloneqq \{x \in \mathbb{R}^{2n} : \|x(t)\| \leq \beta(\|x(0)\|, t) + \alpha(\|u\|_\infty)\}$. Thus, due to the boundedness of the reference trajectories $q_d,\ \dot{q}_d$ and invoking Theorem 1 we conclude that $|\epsilon_{1,i}| \leq |O(\omega\rho_i) + e_i|$ and $|\epsilon_{2,i}| \leq |O(\omega\rho_i) + \dot{e}_i|$. Next, let us define the transformed tracking error $\eta \coloneqq \frac{1}{2}\ln\left(\frac{1+\zeta}{1-\zeta}\right)$ and consider the Lyapunov function candidate $V \coloneqq \frac{1}{2}\eta^2$. Differentiating V with respect to time and substituting (8), yields:

$$\dot{V} = \frac{\eta}{\rho_u(1 - \zeta^2)}\left(\sigma_u^T \dot{\sigma} + c_n(\sigma_n, \rho_u) - k_u \sigma_u^T(u - u_d)\right) \tag{11}$$

with:

$$\dot{\sigma} \coloneqq \begin{cases} \dot{\epsilon}_2 + Q\dot{\epsilon}_1 & \text{if } |\epsilon_2| \leq \bar{\epsilon}_2 \wedge |\epsilon_1| \leq \bar{\epsilon}_1 \\ \dot{\epsilon}_2 & \text{if } |\epsilon_2| \leq \bar{\epsilon}_2 \wedge |\epsilon_1| > \bar{\epsilon}_1 \\ Q\dot{\epsilon}_1 & \text{if } |\epsilon_2| > \bar{\epsilon}_2 \wedge |\epsilon_1| \leq \bar{\epsilon}_1 \\ 0 & \text{if } |\epsilon_2| > \bar{\epsilon}_2 \wedge |\epsilon_1| > \bar{\epsilon}_1 \end{cases}$$

and $c_n(\sigma_n, \rho_u) \coloneqq \sigma_n l\left(1 - \frac{\rho_{u,\infty}}{\rho_u}\right)$.

Based on Theorem 1 in [11], the boundedness of s_n ensures the boundedness of ρ_u. Leveraging the boundedness of $a_i,\ \epsilon_{1,i},\ \epsilon_{2,i},\ e_i, u$ and the fact that $\rho_u \geq \rho_{u,\infty}$ we conclude that there exist a positive constant $\bar{F}$ such that:

$$\left\|\sigma_u^T \dot{\sigma} + c_n(\sigma_n, \rho_u) - k_u \sigma_u^T u\right\|_\infty \leq \bar{F}$$

for all $t \in [0, t_{\max})$. Substituting the control law (6) into (11) and exploiting the fact that $\sigma_u^T \sigma_u = 1$ and $\eta \geq 0$, we obtain:

$$\dot{V} \leq \frac{1}{\rho_u(1 - \zeta^2)}\left(\bar{F}\eta - \frac{k_u \lambda_{\min}(K)}{1 - \zeta^2}\eta^2\right) \tag{12}$$

where $\lambda_{\min}(K)$ denotes the smallest value of the gain matrix K. Since $\frac{1}{1-\zeta^2} > 1$ for all $\zeta \in (-1, 1)$ we conclude that

$\dot{V} \leq 0$ when $\eta \geq \frac{\bar{F}}{k_u \lambda_{\min}(K)}$, which implies that:

$$\eta \leq \bar{\eta} := \max\left\{\eta(0), \frac{\bar{F}}{k_u \lambda_{\min}(K)}\right\}.$$

Taking the inverse logarithmic function we have:

$$0 =: \underline{\zeta} \leq \zeta \leq \bar{\zeta} := \frac{e^{\bar{\eta}} - 1}{e^{\bar{\eta}} + 1} < 1.$$

Hence, $\chi \in \Omega_x \times [\underline{\zeta}, \bar{\zeta}] \subset \mathbb{R}^{2n} \times (-\infty, 1)$ for all $t \in [0, \tau_{\max})$. According to Proposition C.3.6 in [13] (pp. 481), there exists a time instant $t' \in [0, t_{\max})$ at which $\chi(t')$ exits the set $\Omega_x \times [\underline{\zeta}, \bar{\zeta}]$. This, however, contradicts the assumption that $t_{\max} < \infty$. Therefore, we conclude that $t_{\max} = \infty$, which implies that all closed-loop signals are uniformly ultimately bounded and also $\|Q\, e_i(t) + \dot{e}_i(t)\| < \rho_u(t) + O(\omega \rho_{o_i}(t)),\ i = 1, \ldots, n,\ \forall t \geq 0$, completing the proof. ■

Remark 3: The output tracking performance is influenced by the evolution of both the adaptive performance function $\rho_u(t)$, which the controller leverages to balance output performance requirements with hard input constraints, and the evolution of the performance functions $\rho_{o_i}(t),\ i = 1, \ldots, n$. The latter regulates the convergence rate of the estimates from the TD (3), which are subsequently used as inputs to the output-feedback controller (5)-(8).

V. Simulation Results

In this section, we demonstrate the effectiveness of the proposed output feedback scheme on a standard 2-DoF robotic manipulator, governed by the Euler-Lagrange model (1). In this setup, the joint angular positions and velocities are represented by $q = \begin{bmatrix} q_1 & q_2 \end{bmatrix}^T$ and $\dot{q} = \begin{bmatrix} \dot{q}_1 & \dot{q}_2 \end{bmatrix}^T$, respectively, while u denotes the torque applied to the joints, acting as the system's input. Details of the model structure and selected parameters are available in [6].

The desired trajectory is set as $q_{d_1}(t) = 2\sin(1.5t) - 1.5\cos(t)$ and $q_{d_2}(t) = -2\cos(1.5t) + 1.4\cos(t)$. The desired trajectory is set as $q_{d_1}(t) = 2\sin(1.5t) - 1.5\cos(t)$ and $q_{d_2}(t) = -2\cos(1.5t) + 1.4\cos(t)$. The controller was configured using the following parameter values: the gains $k_1 = 45$, $k_2 = 10$, and $k_u = 1$, with adaptation rates $\gamma_1 = 50$, $\gamma_2 = 100$. The control input upper bounds were set to $\bar{u}_1 = 34$ and $\bar{u}_2 = 16$. The rest of the parameters were defined as $\bar{\epsilon}_i = 4$, $q_i = 1.5$, $i = 1, 2$, $l_u = 5$, $l_{o_i} = 5$, $\rho_u^f = 0.01$, $\rho_{o_i}^f = 0.001$ and $\omega = 0.5$. The output performance of the base and upper link of the manipulator, are illustrated in Fig. 1(a),(c) and Fig. 1(b),(d), respectively. The generated torques for the base and upper link of the manipulator, are shown in Fig. 1(e) and Fig. 1(f), respectively. In addition, Fig. 2(a) depicts the evolution of the surface error $\sigma_n(t)$ along with the performance function $\rho_u(t)$. Fig. 2(b) illustrates the performance of the proposed TD for the base and upper link of the manipulator, respectively. It becomes evident that despite only measuring the output tracking error $e(t)$, accurate tracking of the reference trajectory is achieved, as dictated by the parameters of the adopted performance function.

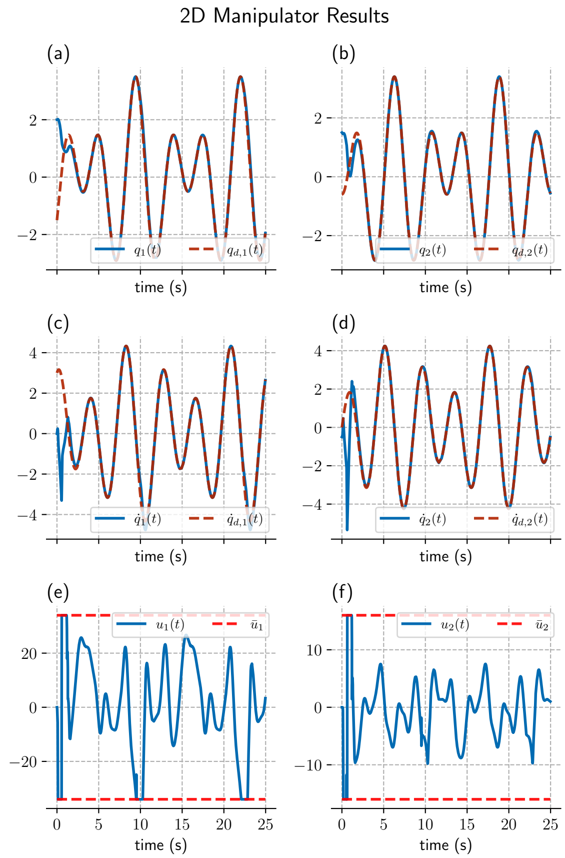

Fig. 1. Output performance and control inputs for the manipulator; (a)-(d): output tracking for the base and upper links; (e)-(f): generated torques.

VI. Experimental Results

In this experiment, the proposed controller was tested on the robotic arm LEAP [16] equipped with dexterous fingers, to autonomously play the piece *"Wind of Change" by Scorpions* on a keyboard, demonstrating the system's precision and control capabilities. The objective was to evaluate the controller's effectiveness in achieving intricate and coordinated multi-fingered movements for musical performance. For this purpose, the control scheme was implemented at the kinematic level, where the control inputs were desired velocities, with the embedded motor controller generating the required torque to achieve them. Consequently, the use of the TD (3) was omitted in this setup, as only the joint position error was provided to the controller (5)-(8). The LEAP Hand has 4 fingers and 16 degrees of freedom, but in this experiment we only use 3 fingers and 12 joints of the hand.

To generate the desired trajectory for the controller, the experiment involved analyzing and extracting the timing and positions of key presses from the musical piece. These extracted data points were used to define target positions for the robotic fingers. To find the desired joint positions and produce dynamically feasible trajectories, we used Trajectory Optimization(TO). The general optimal control problem can be defined as:

$$\underset{x(t),u(t)}{\operatorname{argmin}} \ \mathcal{J}(x(t),u(t)) = \int_{t_0}^{t_f} \ell(x(t),u(t))\,dt + \ell_F(x(t_f))$$
$$\text{s.t.} \quad \dot{x}(t) = f(x(t),u(t)) \tag{13}$$

where

- $x(t) \in \mathbb{R}^{N_s}$, $u(t) \in \mathbb{R}^{N_u}$ are the state and control trajectories
- $\mathcal{J}(x(t),u(t))$ is the "cost function",
- $\ell(x(t),u(t))$ is the "stage cost",
- $\ell_F(x(t_f))$ is the "terminal cost",
- $\dot{x}(t) = f(x(t),u(t))$ are the "dynamics constraints",
- we add more constraints for joint, and actuation limits.

The above formulation results in an infinite-dimensional problem that is generally unsolvable on a computer. Numerous methods exist to transcribe this into a finite-dimensional problem, with discretization being the most straightforward approach [17], [18]. Discretizing the above problem leads to the following generic formulation:

$$\underset{x_{1:K},u_{1:K-1}}{\operatorname{argmin}} \ \mathcal{J}(x_{1:K},u_{1:K-1}) = \sum_{k=1}^{K-1} \ell(x_k,u_k) + \ell_F(x_K)$$
$$\text{s.t.} \quad x_{k+1} = f_{\text{discrete}}(x_k,u_k) \tag{14}$$

where $x_k \in \mathbb{R}^{N_s}$ and $u_k \in \mathbb{R}^{N_u}$. We optimize directly over the knot points. We devise an objective function that aims to minimize velocities and torques, and we add equality constrains for all the desired end effector positions at key times. The constraint gradients require computations of the Jacobians of the end effector frames in the term $\frac{\partial x_t}{\partial q} = J$, where x_t the end effector position and q the joint angles, which are computed through the Pinocchio library [19]. We use the Ipopt [20] solver to optimize the above problem. The optimal solution contains $q, \dot{q}, \tau$ and we use the joint positions to create the reference trajectory.

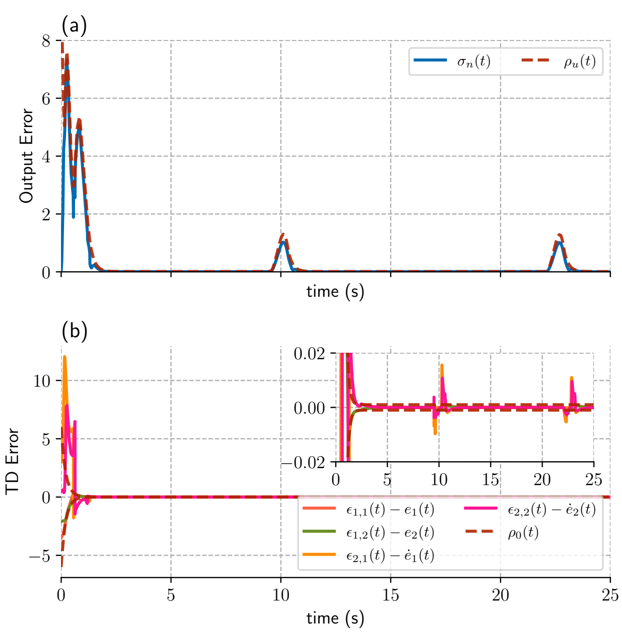

Fig. 2. Error evolution; (a): illustrates the evolution of the surface error $\sigma_n(t)$ along with the performance function $\rho_u(t)$; and (b): performance of the TD (3).

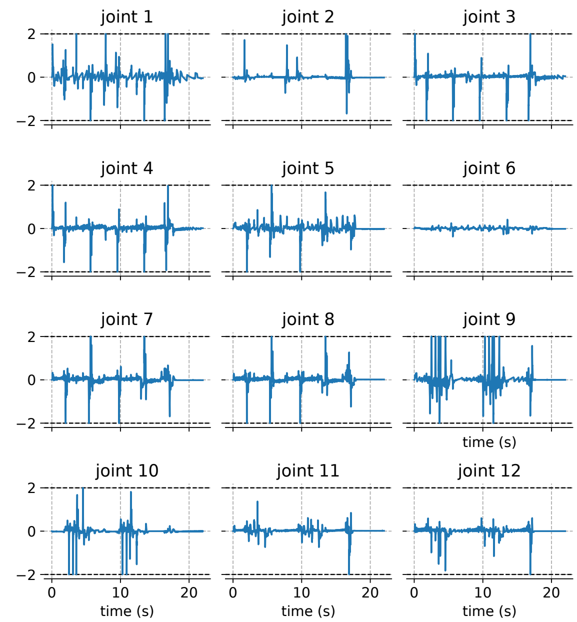

Fig. 3. Joint command velocities for all the joints of the LEAP hand experiment. The dotted lines represent the saturation level.

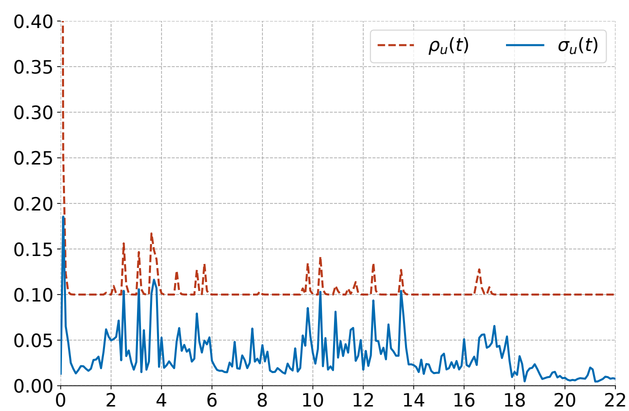

Fig. 4. Adaptive performance funnel $\rho_u(t)$ and the surface error norm $\sigma_n(t)$.

Fig. 3 shows the command velocities for each joint of the first finger, with dotted lines indicating saturation levels. The controller successfully maintains velocities within these limits, highlighting its effectiveness in imposing adaptive performance constraints in real-time.

Fig. 4 illustrates the adaptive performance funnel $\rho_u(t)$ and the surface error norm $\sigma_n(t)$, while Fig. 5 depicts joint position tracking performance, across all joints, with each line corresponding to a different finger. Fig. 5 shows that each joint accurately follows its reference trajectory, with red lines representing the desired positions. The performance

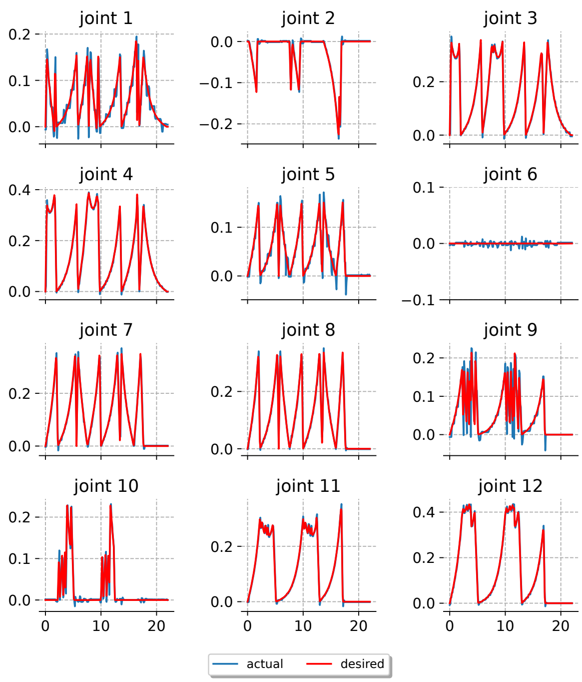

Fig. 5. The joint positions of all joints of the LEAP hand experiment. Red lines show the desired positions, and blue lines the actual positions.

funnel adapts dynamically, ensuring the tracking error remains within prescribed bounds, confirming the controller's capability to enforce precise position tracking validating the robustness and precision of the proposed control scheme for this complex, multi-fingered robotic task. The experimental results are available in https://www.youtube.com/watch?v=Mv-cQfQ25PI.

VII. Conclusions

This work has demonstrated that a single-funnel APC controller can ensure robust output tracking in high-DoF robotic manipulators. The proposed control scheme imposes adaptive performance specifications for both joint position and velocity, accommodating hard actuation constraints while maintaining the boundedness of all closed-loop signals. The robust tracking differentiator further enhances control applicability by accurately estimating unavailable velocity information, thus reducing dependence on velocity sensors or prior knowledge of reference velocity. Simulation results on a planar 2-DoF manipulator, alongside experimental validation on a 3-fingered, 12-DoF robotic hand, confirm the efficacy of the controller in real-world scenarios and its robustness in tasks requiring precise control under physical constraints. Future work will focus on extending this approach to more complex, task-specific applications.

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